

Multi-Hop Biomedical Reasoning

Retrieval-Augmented Generation for Drug Interaction QA Using Local LLMs

Presented by: Mariam Aadel Abdulaal

Supervised by: Dr. Riad Sonbol | Eng. Aya Alaswad

Institution: Syrian Private University

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Bachelor's Degree Defense

QAnGaroo MedHop v1.1

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Problem Statement & Dataset

The Research Problem

- ▶ **Task:** Given a query drug and a candidate list, identify which candidate drug interacts with it through multi-hop reasoning across PubMed abstracts and DrugBank.
- ▶ **Challenge:** A single document rarely contains complete evidence. The system must chain facts through intermediate bridge entities across multiple sources.
- ▶ **Constraint:** Fully local inference only — no paid APIs, no fine-tuning. RTX 5050 laptop, 8 GB VRAM, 4-bit quantized LLMs (7B–9B parameters).
- ▶ **Metric:** Exact Match (EM) — prediction must exactly match the DrugBank ID of the correct answer drug.

QAnGaroo MedHop v1.1

342

Dev Questions
(evaluation set)

1,620

Train Questions
(classifier training)

~8.6

Avg Candidates
per Question

133

Unique
Answer Drugs

11.63%

Random Baseline
(EM)

76.9%

Oracle Upper Bound
(gold chain k=3)

Research Approach — Six Development Stages

1

Baseline 1: Direct LLM QA

BioMistral-7B — parametric knowledge only, no retrieval

15.8%

2

Baselines 2–3: Sparse & Dense RAG

BM25 (sparse) + MedCPT (dense) retrieval with BioMistral-7B

16.4%

3

Stage 3: Hybrid + Qwen3.5-9B

Hybrid Scored Retrieval (BM25+MedCPT) + Guided Prompt at k=3

33.3%

4

Stage 4: Entity Bridging

LLM extracts bridge drug → bridge-boosted retrieval → answer

33.3%

5

Stage 5: Ensemble — Majority Vote

3-way vote: Stage 3 + Entity Bridge + Ontology Verification

35.4%

6

Stage 6: Meta-Classifer ★ BEST

Logistic Regression on 1,976 features — selects best path per question

51.17%

Baseline Pipeline — How Baseline 1 Works (Direct LLM QA)

Result: 15.8% EM (54/342)

INPUT

Query Question

"Which drug interacts with DrugX through mechanism Y?"

Candidate drug list:
[DB00001, DB00002, ...]

PROCESS

BioMistral-7B (no retrieval)

Model uses only its internal parametric knowledge.

No documents retrieved.
No external data.

Generates answer directly from training memory.

OUTPUT

Predicted DrugBank ID

"DB00441"

Compared against gold answer by Exact Match.

Result: 15.8% EM
(54/342 correct)

⚠ Why it fails: multi-hop drug interactions require chaining evidence across multiple documents — parametric knowledge alone is insufficient (15.8% EM vs 76.9% oracle)

Entity Bridging — How Best Experiment (Stage 4) Works

Result: 33.3% EM (114/342)

STEP 1

Bridge Extraction

- Qwen3.5-9B uses parametric knowledge
- Identifies intermediate bridge drug
- Example: Drug A → Bridge → Drug B
- No documents retrieved at this stage

Bridge entity identified

STEP 2

Bridge-Boosted Retrieval

- Hybrid retrieval: MedCPT + BM25
- Bridge drug name injected as boost signal
- $\text{Score} = 0.5 \times \text{MedCPT} + 0.5 \times \text{TermCount} + w \times \text{bridge}$
- Top k=3 documents selected (optimal)

3 relevant docs retrieved

STEP 3

Answer Generation

- CoT Enriched guided prompt
- Incorporates retrieved docs + bridge hint
- Qwen3.5-9B selects from candidate list
- Outputs DrugBank ID as final answer

33.3% EM (114/342)

Ensemble — How the 3-Way Majority Vote Works

INPUT — Query Drug + Candidate List

Result: 35.4% EM (121/342)

Path 1

Stage 3 (Hybrid)

Hybrid Retrieval
+ Qwen3.5-9B
(Guided Prompt)

33.3% individual EM

Path 2

Stage 4 (Entity Bridge)

Bridge-Boosted
Retrieval
+ Qwen3.5-9B

33.3% individual EM

Path 3

Stage 4 (OV-EXP4)

Ontology
Verification
+ Qwen3.5-9B

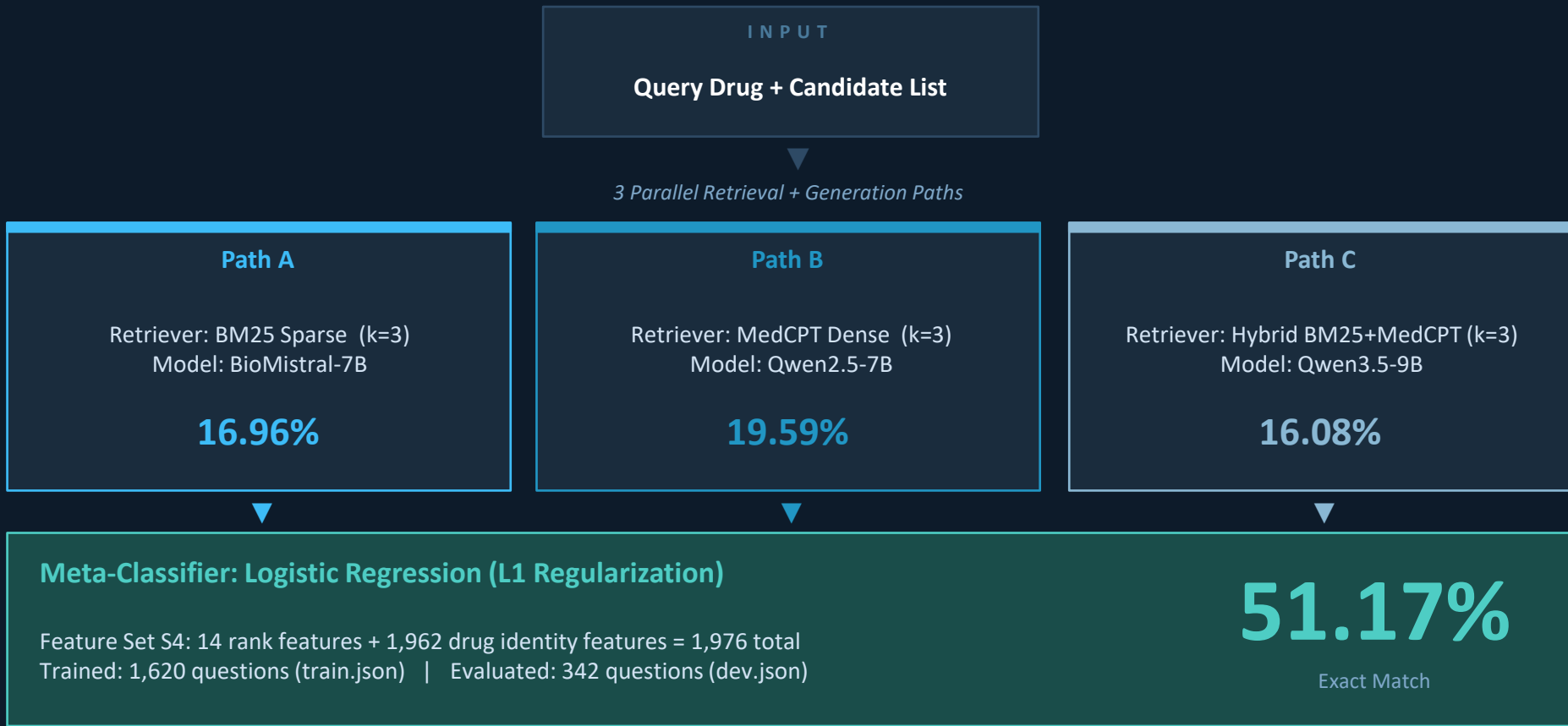
32.2% individual EM

Majority Vote — Each question: the candidate predicted by ≥ 2 paths wins; if no majority, Path 2 (best individual) decides

35.4%

Ensemble EM

★ Best System Architecture — Meta-Classifer Pipeline (51.17% EM)



Meta-Classifer — Feature Engineering Design

Feature Set S4 — 1,976 total features (best configuration | AUC 0.8054 | AUC Gap 0.036 — minimal overfitting)

Drug Identity Features (1,962 features — 99.3% of total)

One-hot encoding of each candidate DrugBank ID.
The classifier learns which specific drugs are consistently answered correctly across all 3 paths.
Captures drug-pair specificity of interactions.

13 of 15 Top Features

are drug identity embeddings

The classifier learns which DRUG is correct
— not which retrieval path is better.

Rank Features (14 features — 0.7% of total)

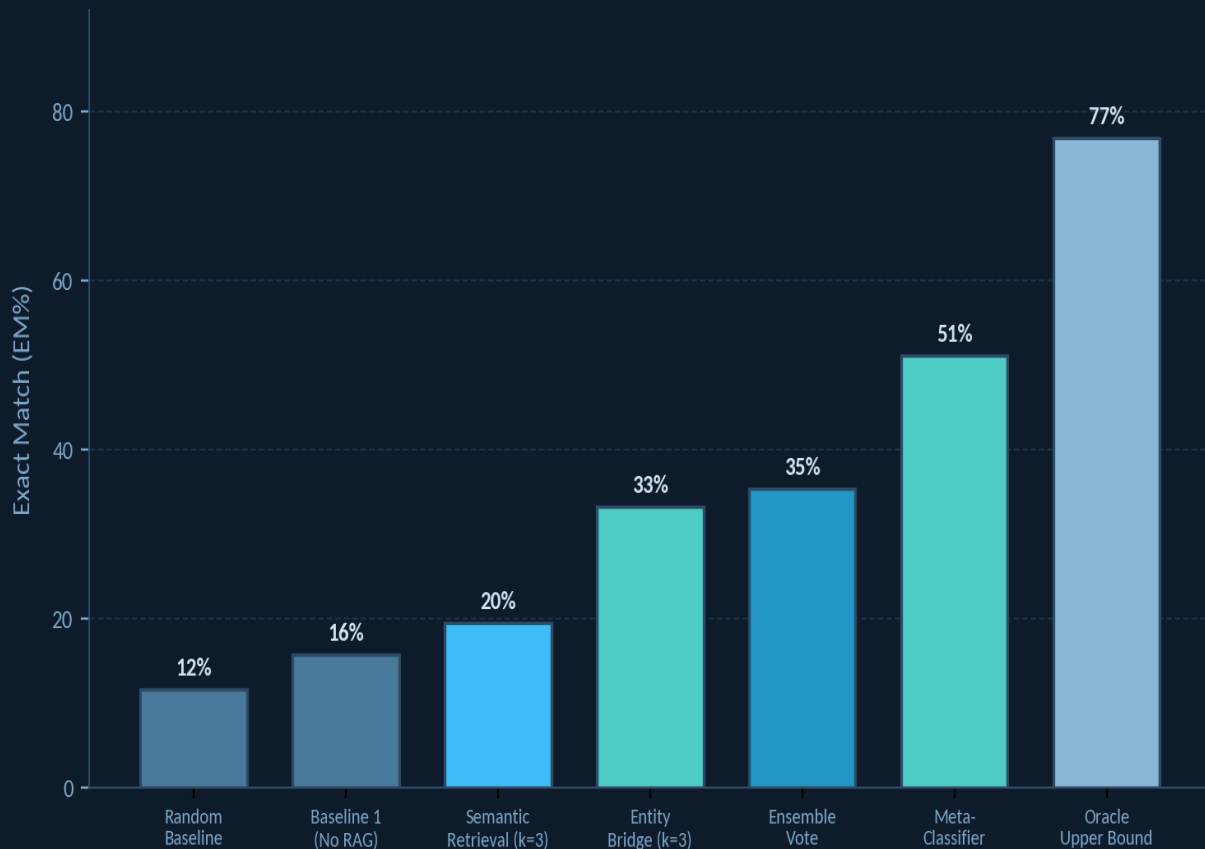
Rank positions (1–9) of candidate in each path
(Path A / B / C), plus recall@K scores per path.
Captures retrieval signal across 3 methods.
14 features total.

Classifier Design Choices

- ✓ Logistic Regression — L1 regularization, $C = 0.01$
- ✓ No data leakage: trained on train.json, tested on dev.json
- ✓ AUC 0.8054 | AUC Gap 0.0357 (minimal overfitting)
- ✓ L1 sparsity: most drug features shrink to 0 — only strongly predictive drugs remain active

Drug interactions are highly drug-pair specific — e.g. Gemcitabine (DB00441) is almost always predicted correctly, regardless of retrieval ranking

Experiments & Results — Stage-by-Stage Progression



15.8%

Direct LLM (no RAG)

Parametric knowledge alone insufficient

33.3%

Entity Bridge (best single)

+17.5pp vs best baseline — Qwen3.5-9B key

35.4%

Ensemble Vote

+2.1pp from 3 complementary paths

51.17%

Meta-Classifier ★

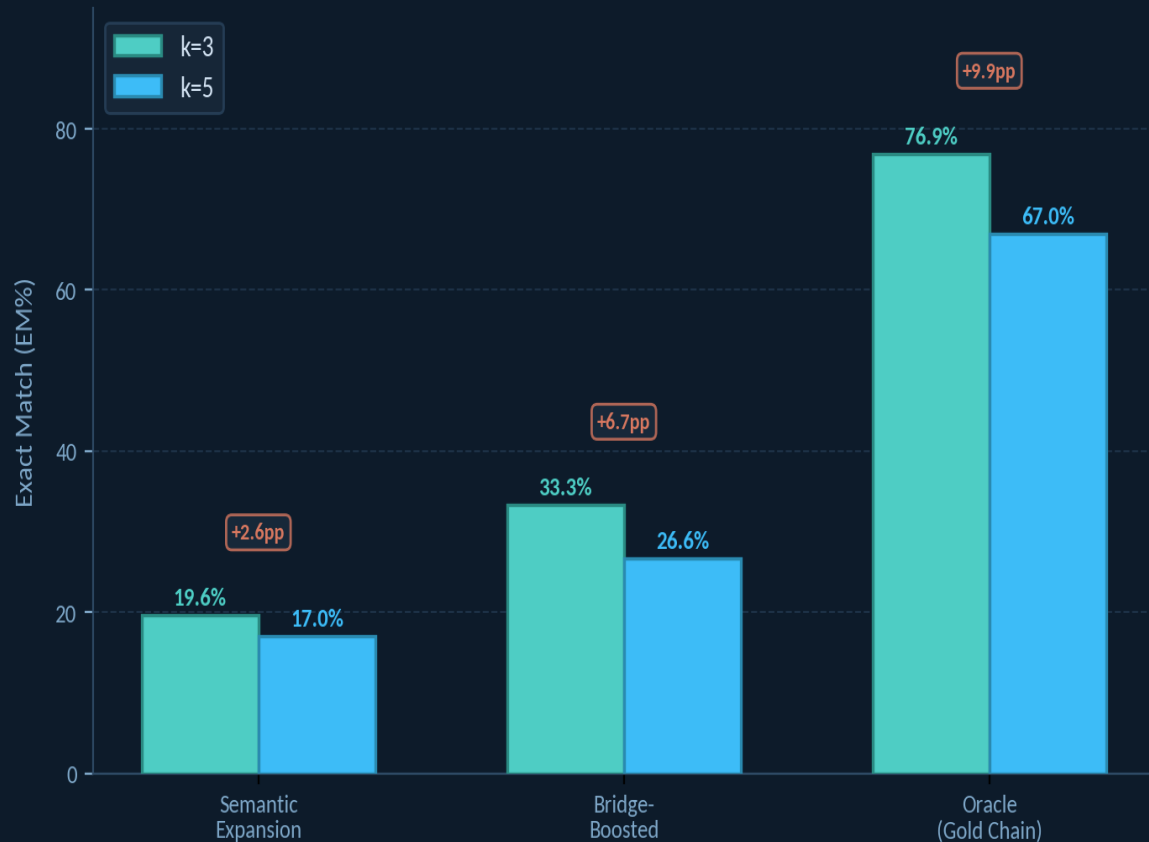
+15.8pp over ensemble — BEST

76.9%

Oracle upper bound

Gold chain k=3 — not feasible in production

Retrieval Depth Analysis — Why k=3 Is Optimal



Best Result @ k=3

33.3% EM

Qwen3.5-9B — All Retrieval Methods

↓ **k=3 consistently outperforms k=5**

Even Oracle (gold chain) performs worse at k=5 (66.96%) vs k=3 (76.90%). More context = noise.

⚠ **Small models, short context**

7–9B models struggle to extract answers from long concatenated documents.

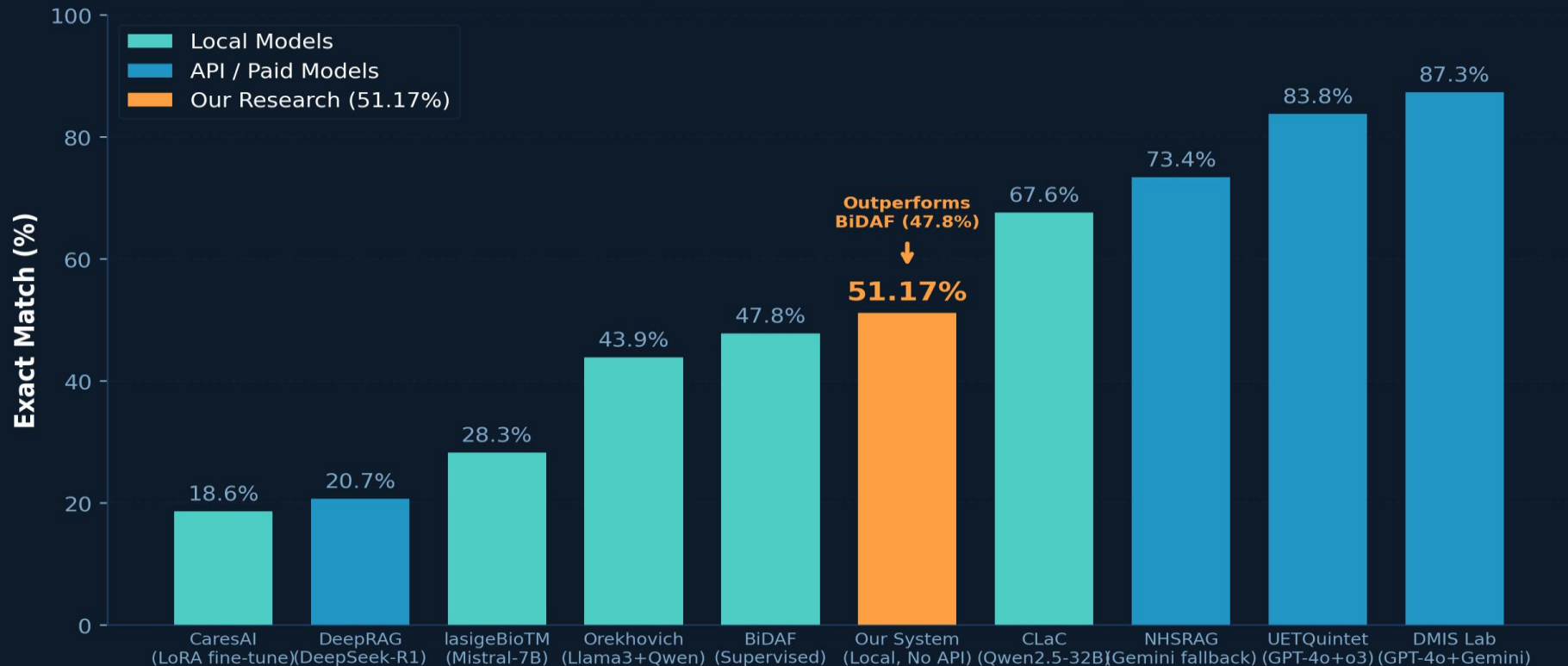
★ **k=3 consistently optimal**

k=3 wins for ALL methods: Hybrid (+4.97pp), Bridge (+6.72pp), even Oracle (+9.94pp).

Published Systems — EM Score Comparison

EM% scores: 9 published MedHop systems + Our System (51.17%). Teal = local models, Blue = API models, Orange = our research.

Published Systems — EM Score Comparison



Comparison with Published Systems

System	Model	EM%	Setup
DMIS Lab (Rank 1) (2025)	GPT-4o + GPT-o3	87.3	3-Arm Ensemble + Web Search
UETQuintet (Rank 2) (2025)	GPT-4o-mini + o3-mini	83.8	Hop-logic classifier + API
NHSRAG (2025)	MedReason-8B + Gemini	73.4	NER + sparse search
CLaC Systems (2025)	Qwen2.5-32B (local)	67.6	Wikipedia retriever + vLLM
Our Best (Local) ★ (2026)	Qwen3.5-9B (7–9B local)	51.17	Meta-Classifer + Ollama (no API)
BiDAF (Welbl 2018)	LSTM ~50M params	47.8	Extractive RC on superdocument
Orekhovich / IBMC (2025)	Llama3-Med42 + Qwen2.5-32B	43.9	Hybrid BM25S + MedEmbed + Ollama
IasigeBioTM (2025)	Mistral-7B + MedGemma-27B	28.3	NER + Mondo Ontology
DeepRAG / BioHop (2025)	DeepSeek-R1 (671B)	20.7	Hierarchical reasoning + DPO
CaresAI (2025)	LLaMA-3 8B (LoRA fine-tuned)	18.6	SFT on BioASQ + MedQuAD

Our Differentiation — Why This Work Matters

What Top Systems Have (We Don't)

- ▶ GPT-4o / GPT-o3 API — expensive per-call
- ▶ Web search during inference (live internet)
- ▶ 32B+ parameter multi-GPU servers
- ▶ Fine-tuning on MedHop-specific data

Our Constraints (By Design)

- ▶ RTX 5050 laptop, 8 GB VRAM only
- ▶ No paid APIs — 100% local via Ollama
- ▶ 4-bit quantized 7–9B models only
- ▶ No task-specific fine-tuning — zero-shot

Our Unique Contributions

- ▶ Meta-classifier: learns which path to trust per question → 51.17%
- ▶ First systematic k / retriever / LLM ablation on MedHop
- ▶ MedCPT + BM25 hybrid + bridge entity boosting (novel combination)
- ▶ Surpasses BiDAF + 3 BioCreative IX systems without fine-tuning

Key Findings & Contributions

Finding 1: Model Over Architecture

With BioMistral-7B, adding retrieval hurt. The model was the bottleneck. Switching to Qwen3.5-9B alone gained +16.9pp.

Finding 2: k=3 Is Optimal

Top-3 documents consistently beat top-5 and top-10. Extra context adds noise — small local models struggle with long inputs.

Finding 3: Simplicity Wins

Complex modules (Query Decomposition, Ontology, Entity Bridging variants) do not surpass well-tuned simple retrieval at 7–9B scale.

Finding 4: Meta-Learning Power

Meta-classifier (51.17%) outperforms voting ensemble (35.4%) by +15.8pp. Trusting the right path per question is far more effective.

Finding 5: Drug Identity Dominates

13 of 15 top features are drug-specific embeddings. The classifier learns which drug is correct — not just where it ranks.

Conclusion & Future Work

Conclusion

- ✓ Systematic study of multi-hop biomedical QA with fully local 7–9B models on consumer hardware (RTX 5050, 8 GB VRAM)
- ✓ Best result: 51.17% EM via Meta-Classifier — outperforms original BiDAF baseline by +3.37pp without any fine-tuning
- ✓ Surpasses 4 published BioCreative IX systems (Orekhovich 43.9%, lasigBioTM 28.3%, CaresAI 18.6%, BiDAF 47.8%)
- ✓ Retrieval quality (not reasoning complexity) is the primary bottleneck — 25.73pp gap to oracle remains
- ✓ Meta-learning (51.17%) surpasses majority voting (35.4%) by +15.8pp — complementary paths matter

Future Work

- Improve retrieval to close the 25.73pp gap to oracle upper bound
- Explore larger quantized models (70B Q4) for reasoning improvements
- Apply meta-classifier framework to other multi-hop QA benchmarks
- Graph-based reasoning over drug interaction networks (PPI chains)
- Online learning: update classifier as new predictions arrive

Thank You

Questions & Discussion

17

Total Slides

33.3%

Best Single Model
(Entity Bridge)

51.17%

Best Result
(Meta-Classifer)

+3.37pp

vs BiDAF Baseline

GitHub: [Mariam6600/Multi-Hop-Biomedical-Reasoning](https://github.com/Mariam6600/Multi-Hop-Biomedical-Reasoning)

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